✈ DGCA CPL / ATPL STUDY NOTES

CHAPTER 4

Antennae

Radio Navigation & Aids — Ground Training Series

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1. Introduction

> **What this section covers:** What antennae do, why the choice of antenna type matters, and what determines which antenna is used in each radio system.

Antennae (or *aerials*) are the means by which radio energy is radiated and received. Every radio system — from a cockpit VHF radio to a precision ILS array — uses an antenna specifically designed for its function.

> **Key Principle**
>
> The type of antenna used is determined by: (1) the frequency in use, (2) whether directional or omnidirectional radiation is required, and (3) the platform (aircraft, ground, ship). This chapter covers the principles common to all antennae, plus specialist types for navigation systems.

2. Basic Principles — Dipole & Marconi Aerials

> **What this section covers:** The two fundamental antenna types for basic communications — the half-wave dipole and the quarter-wave Marconi (which is used on aircraft).

There are two basic aerial types for communications:

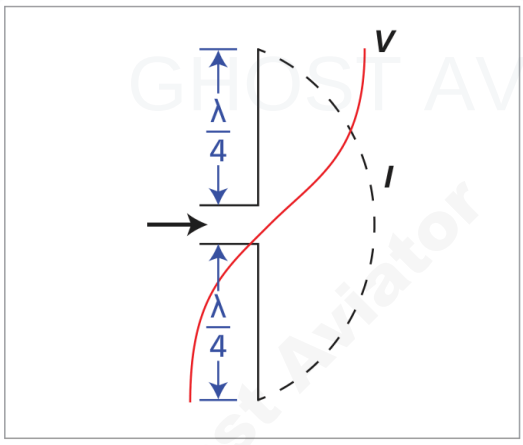


Figure 4.1: Half-wave dipole

Figure 4.1 — Half-wave dipole: power fed to the centre; two λ/4 elements extend in opposite directions. Current (I) is maximum at the

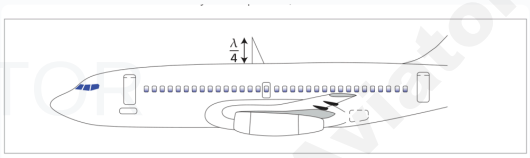


Figure 4.2: Marconi aerial

Figure 4.2 — Marconi (quarter-wave) aerial: one λ/4 element above a conductive ground plane, which acts as the second half of a dipole via reflection. Used on aircraft for better aerodynamics

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centre, voltage (V) maximum at the tips. Radiates perpendicular to the aerial axis

Feature	Half-Wave Dipole	Quarter-Wave Marconi
Total aerial length	λ/2	λ/4
Ground plane	Not required	Required (metal surface)
Radiation direction	Perpendicular to aerial	Perpendicular to aerial
Aircraft use	Not ideal (aerodynamics)	Standard on aircraft (blade aerial)
How it works	Centre-fed, both halves radiate	Ground plane acts as second half of dipole

💡 **Why Marconi Aerials on Aircraft?**

The Marconi aerial's single-element design has **better aerodynamic qualities** (blade aerial on aircraft fuselage). The metal aircraft skin acts as the ground plane. Dipoles need two exposed elements, which would create aerodynamic drag and structural problems at aircraft speeds.

3. Aerial Length & Velocity Factor

What this section covers: The wavelength formula, theoretical aerial lengths, the velocity factor correction, and the key calculation type for exam questions.

For maximum efficiency, an aerial must be the **correct length** for the wavelength of the frequency in use — ideally λ/2 (dipole) or λ/4 (Marconi).

WAVELENGTH FORMULA (FROM CHAPTER 1)

```
λ (metres) = 300 / f (MHz)

Dipole length = λ / 2

Marconi length = λ / 4
```

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Velocity Factor — Why the Actual Aerial Is Shorter

EM energy travels at the speed of light *in free space*. In a physical conductor (the aerial element), the speed is slightly slower — the denser medium reduces propagation velocity by approximately **5%**. This means the electrical wavelength inside the conductor is shorter than the free-space wavelength.

Therefore, the **actual optimum aerial length** is approximately **95%** of the theoretical value:

ACTUAL AERIAL LENGTH WITH VELOCITY FACTOR

```
Actual length = 0.95 × Theoretical length

Actual Marconi = 0.95 × (λ/4)

Actual Dipole = 0.95 × (λ/2)
```

📐 Worked Examples — Aerial Length Calculations

Example 1 (textbook): Marconi aerial at 125 MHz

λ = 300/125 = 2.4 m

Theoretical λ/4 = 2.4/4 = 0.6 m = 60 cm

Actual = 0.95 × 60 cm = 57 cm

Example 2 (Q1 in chapter): Marconi at 406 MHz

λ = 300/406 = 0.739 m

Theoretical λ/4 = 0.739/4 = 0.1847 m = 18.5 cm

Actual = 0.95 × 18.5 = **17.6 cm ≈ 17.5 cm** ✓ (answer c)

Example 3 (Q4 in chapter): Dipole at 75 MHz

λ = 300/75 = 4.0 m

Theoretical λ/2 = 4.0/2 = 2.0 m

Actual = 0.95 × 2.0 = **1.9 m** ✓ (answer a)

4. Aerial Feeders

> **What this section covers:** How energy travels between the aerial and the transmitter/receiver, and why different feeder types are needed at different frequency

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bands.

The means of carrying RF energy between the aerial and equipment depends on **frequency and power level**. As frequency increases, losses in simple feeders become unacceptable.

Frequency Band	Feeder Type	Why
LF & MF	Simple wire	Low losses at low frequency over reasonable distances
HF & VHF	Twin wire feeder	Single wire losses too high; twin wire more efficient
UHF	Coaxial cable	Twin wire losses unacceptably high; coax required
Upper UHF, SHF, EHF	Waveguide	Coax losses impractical; dipole/Marconi aerials not usable at these frequencies

🔵 **Waveguide — Key Facts**

- A **hollow, rectangular metal tube** — no inner conductor
- Internal dimensions determined by frequency: internal dimension = **λ/2** (half the wavelength)
- Used in radar systems (UHF/SHF) to carry high-power pulses from transmitter to the parabolic dish or phased array antenna
- At SHF and EHF, dipole and Marconi aerials are impractical — the waveguide itself acts as the radiating element (horn feed)

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5. Polar Diagrams

What this section covers: How to read and interpret polar diagrams, the radiation pattern of a dipole aerial, and the torus 3-D representation.

A **polar diagram** shows the radiation (or reception) pattern of an aerial — it is a line joining all points of equal signal strength, generally shown as a plan view perpendicular to the plane of radiation. (The same polar diagram applies to both transmission and reception — aerials are reciprocal.)

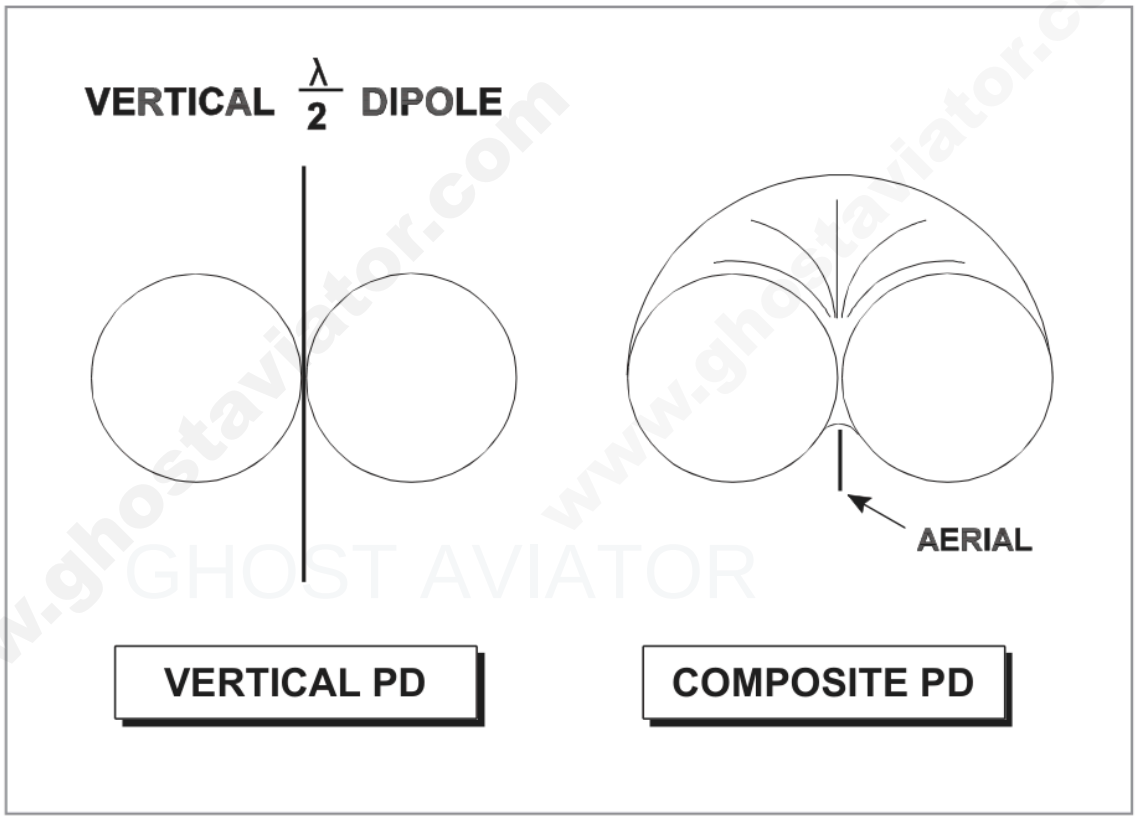


Figure 4.3 3-D Polar Diagram (PD)

Figure 4.3 — 3-D Polar Diagram (torus): a dipole aerial radiates maximum energy perpendicular to its axis. The 3-D pattern is a torus (doughnut shape) centred on the midpoint of the aerial. The Vertical PD (right) and Composite PD (left) are shown

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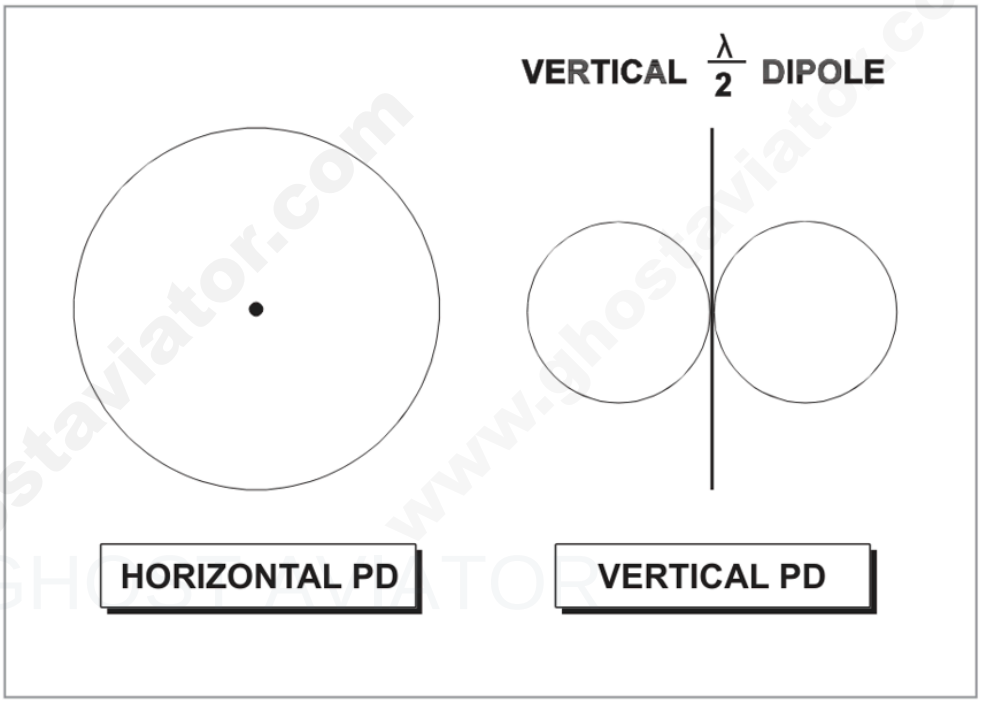


Figure 4.4 Plan view polar diagram

Figure 4.4 — Plan view polar diagrams: the horizontal PD of a vertical dipole is a circle (omnidirectional in the horizontal plane); the vertical PD shows the torus cross-section with nulls at the aerial tips

🔵 **Dipole Radiation Pattern**

- Maximum radiation: **perpendicular (90°) to the aerial**
- Minimum radiation (nulls): **at the tips of the aerial** (along the aerial axis)
- 3-D pattern: a **torus** (doughnut shape) — symmetrical around the aerial axis
- Horizontal plan view of a vertical dipole: a **perfect circle** (omnidirectional)

6. Directivity

What this section covers: How directivity is achieved using parasitic elements (reflectors and directors), the side lobe problem, ILS localizer array design, and the ADF loop aerial.

6.1 Reflectors & Directors (Yagi Array)

Many navigation systems require **directional** emission/reception. The simplest method is adding **parasitic elements** to the basic aerial:

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Reflector — How It Works

- A metal rod **5% longer** than the aerial
- Placed at a distance of **λ/4** behind the aerial, in the same plane
- Re-radiates energy **180° out of phase** → cancels signal behind, reinforces signal in front
- Result: no radiation behind the reflector; increased forward gain

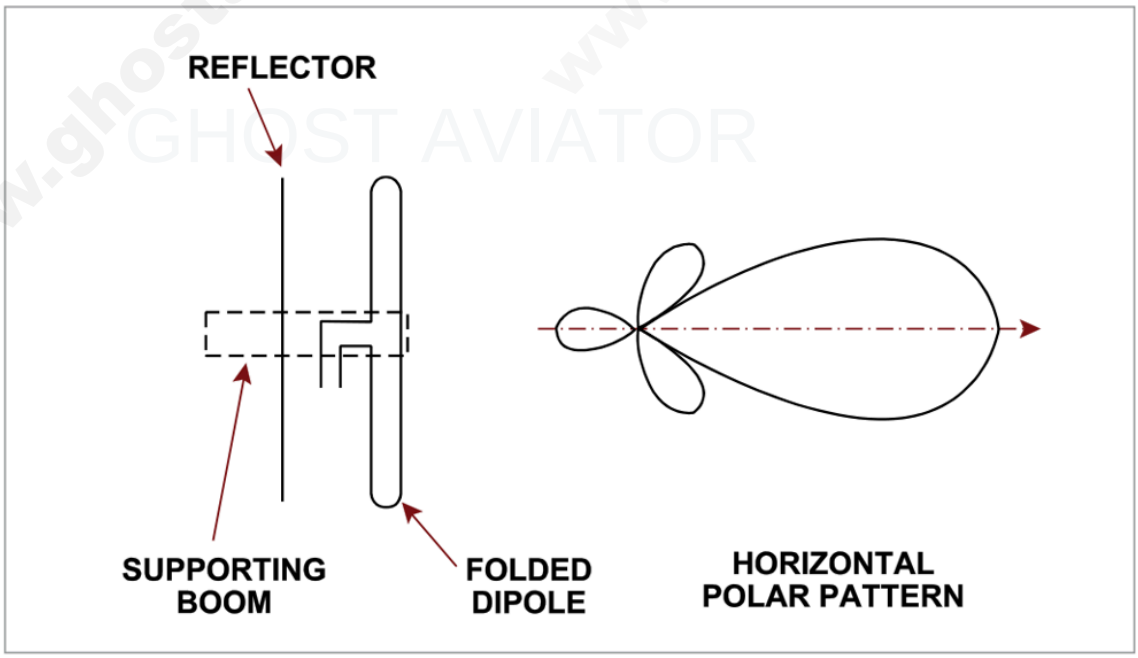


Figure 4.5 Directivity using reflector

Figure 4.5 — Directivity using reflector: one parasitic reflector element (longer than the aerial, at λ/4 spacing) produces a cardioid-type pattern with maximum gain forward and a null behind. The polar diagram shows the beam now concentrating forward

Directors — How They Work

- Metal rods **shorter** than the aerial
- Placed **in front** of the aerial (toward the desired direction)
- Act as focusing elements — concentrate the signal into (or out of) the aerial
- Multiple directors produce progressively narrower, stronger beams

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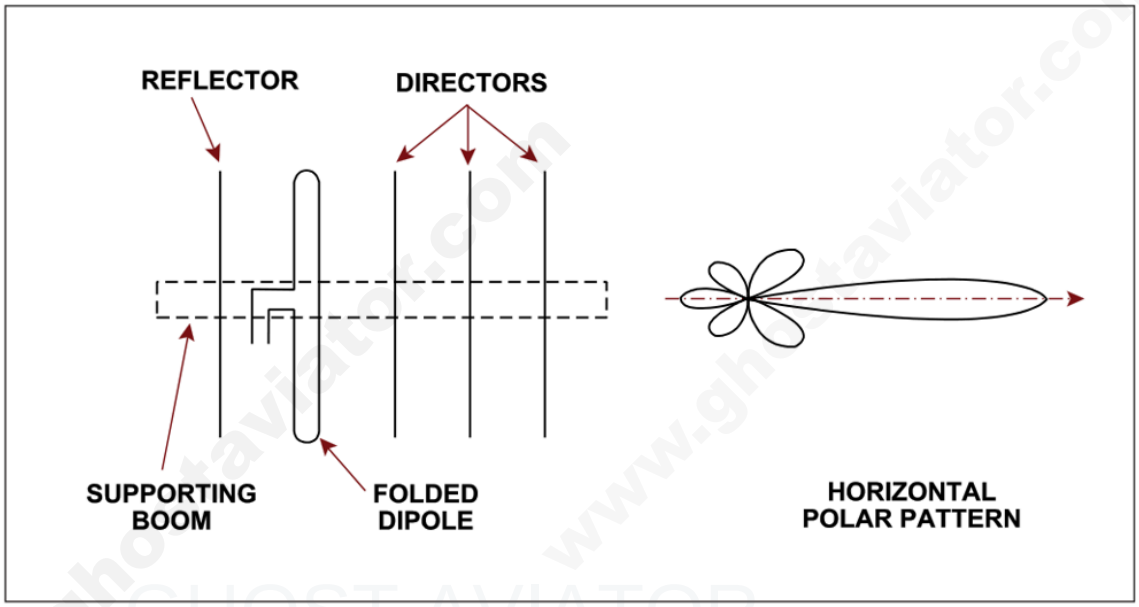


Figure 4.6 Improved directivity using reflector and directors

Figure 4.6 — Improved directivity using one reflector and multiple directors: the array focuses energy into a narrow forward beam (maximum gain). This is the classic Yagi antenna design — recognizable as a TV aerial array

> ⚠ **The Side Lobe Problem**
>
> While a narrow main beam is produced, **unwanted side lobes** also appear. These side lobes:
>
> - Receive and transmit signals in unintended directions
> - Cause TV ghosting (reflections from buildings)
> - Create significant problems in **SSR** and **ILS**
> - Produce spurious returns in primary radar systems
>
> Side lobe suppression is a major engineering challenge in aviation antenna design.

6.2 ILS Localizer Array

The ILS uses an extension of the Yagi array principle to produce the precision beams required to guide aircraft to the runway. The **ILS localizer antenna** is:

- An array of **16 or 24 aerials** placed in a line
- Spacing of **λ/2** between each aerial
- Produces **two narrow symmetrical lobes** either side of the runway centreline

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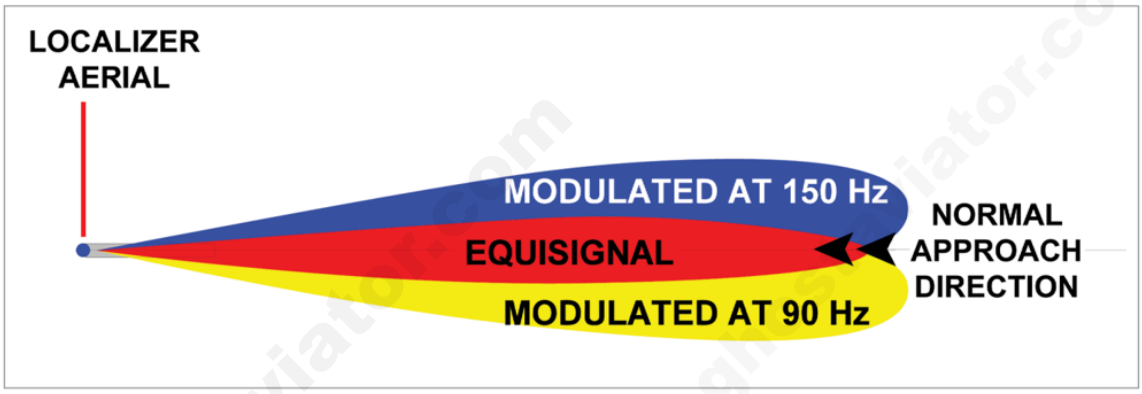


Figure 4.7 ILS localizer lobes.

Figure 4.7 — ILS localizer beam pattern: two narrow symmetrical lobes close to the runway centreline are produced by the 16- or 24-element array. An aircraft deviating from centreline receives unequal signals from the two lobes — this difference drives the course deviation indicator

6.3 ADF Loop Aerial

The **ADF (Automatic Direction Finder)** uses a **loop aerial** to determine the direction of an incoming radio signal.

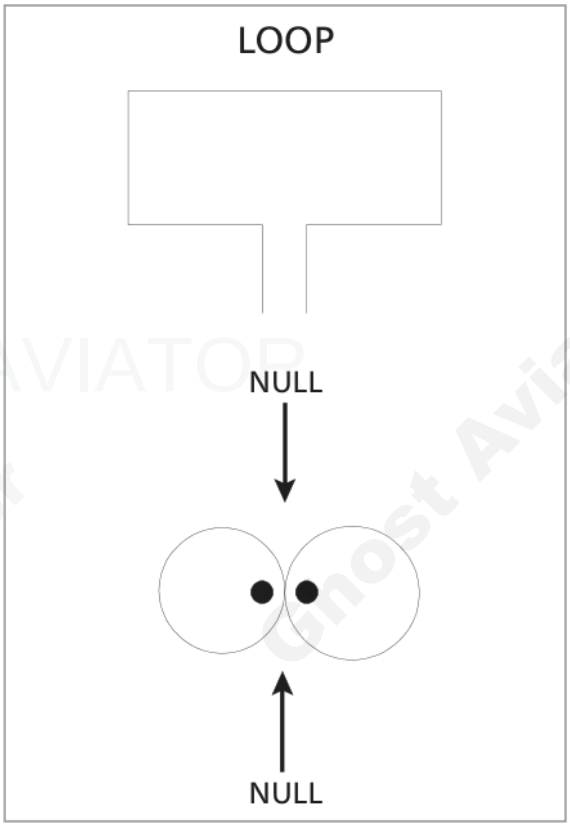


Figure 4.8 Loop aerial 'Figure-of-eight' polar diagram

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Figure 4.8 — ADF loop aerial: the figure-of-eight polar diagram has two distinct nulls 180° apart (when the loop plane is at 90° to the incoming signal). Maximum sensitivity when the loop plane aligns with the incoming signal; the two nulls are used to determine bearing

🔵 **How the Loop Aerial Works — Phase Difference Principle**

- **Loop aligned with signal**: phase difference between signals in the two vertical elements → **net current flows** → signal received (maximum)
- **Loop at 90° to signal**: induced currents in both vertical elements are equal and opposite → cancel out → **zero output (null)**
- Result: a **figure-of-eight polar diagram** with **two nulls**
- The two nulls are used to determine the bearing to the NDB transmitter

💡 **Why Two Nulls Are Better Than Two Maxima**

The null is used for direction finding (rather than the maximum) because the null is much **sharper** — a small angular change produces a large signal change. This gives better bearing accuracy. The loop aerial has two nulls 180° apart, which causes a **180° ambiguity** — resolved by combining with a sense aerial (discussed in Chapter 7 — ADF).

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7. Radar Aerials — Parabolic Dish

What this section covers: How the parabolic reflector works, the role of the waveguide horn feed, the focal point principle, and the problem of side lobes from the dish.

Radar systems operate in **UHF and SHF bands** — requiring waveguides rather than cables. The parabolic dish is the classic radar antenna:

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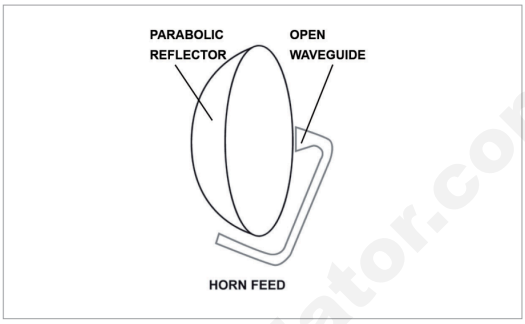


Figure 4.9 Horn feed to Parabolic Reflector

Figure 4.9 — Horn feed: the open end of a waveguide (horn) is positioned at the focal point of the parabolic dish. RF energy radiates from the horn and is reflected by the dish

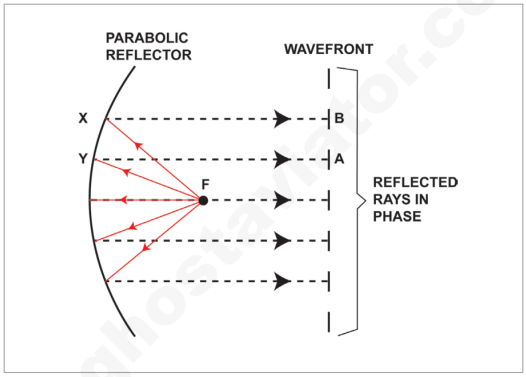


Figure 4.10 Principles of the Parabolic Reflector

Figure 4.10 — Focal point principle: paths FXB and FYA are equal — all reflected rays travel the same distance → arrive in phase → form a uniform plane wavefront

How the Parabolic Dish Works

- Open end of waveguide (horn) placed at the **focal point (F)** of the parabola
- RF energy from horn is reflected by dish as **parallel rays**
- Path length from focal point to dish to output is **equal for all rays** → all reflected rays are **in phase**
- This creates a **uniform plane wavefront** — narrow pencil beam
- In practice the beam diverges beyond the near field — parallel only close to the antenna

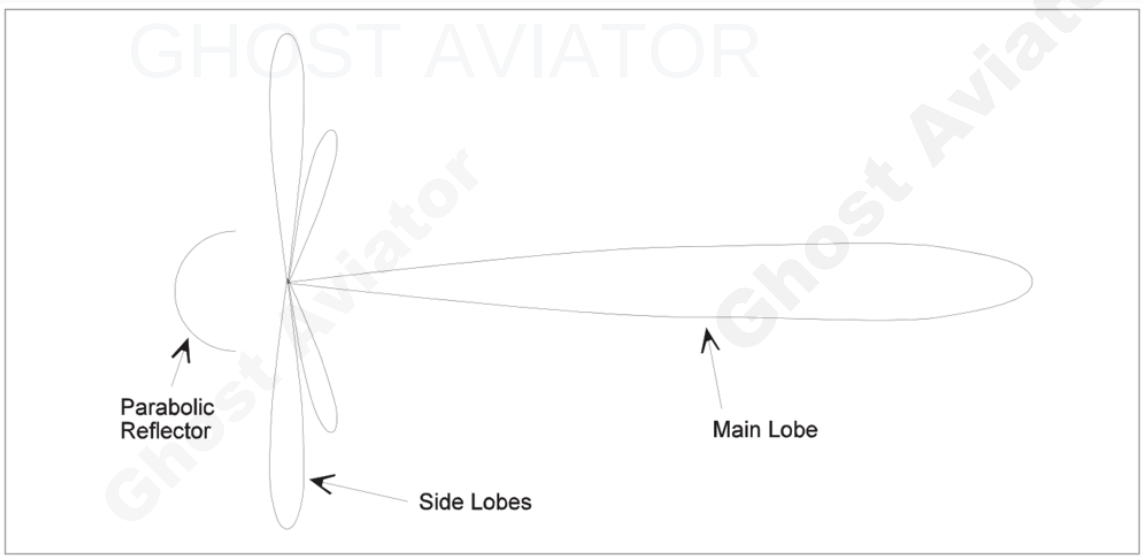


Figure 4.11 Polar diagram of parabolic reflector

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Figure 4.11 — Polar diagram of parabolic reflector: a narrow main lobe plus multiple side lobes from energy that 'spills' beyond the edges of the reflector dish. Side lobes contain enough energy to produce false radar returns

> ⚠ **Side Lobe Problem — Parabolic Dish**
>
> Due to **uneven reflection** and energy spillage around the edges of the dish, side lobes are produced. These side lobes:
>
> - Contain sufficient energy to produce valid radar returns outside the main beam
> - Can cause **false targets** on radar displays
> - Are addressed in modern radar by switching to phased array technology

8. Modern Radar Antennae — Phased Array

> **What this section covers:** The flat plate / phased array / slotted antenna design, how it works, and its five advantages over the parabolic dish.

Modern radar development has replaced the parabolic dish with the **Flat Plate Array** (also called *Phased Array* or *Slotted Antenna*).

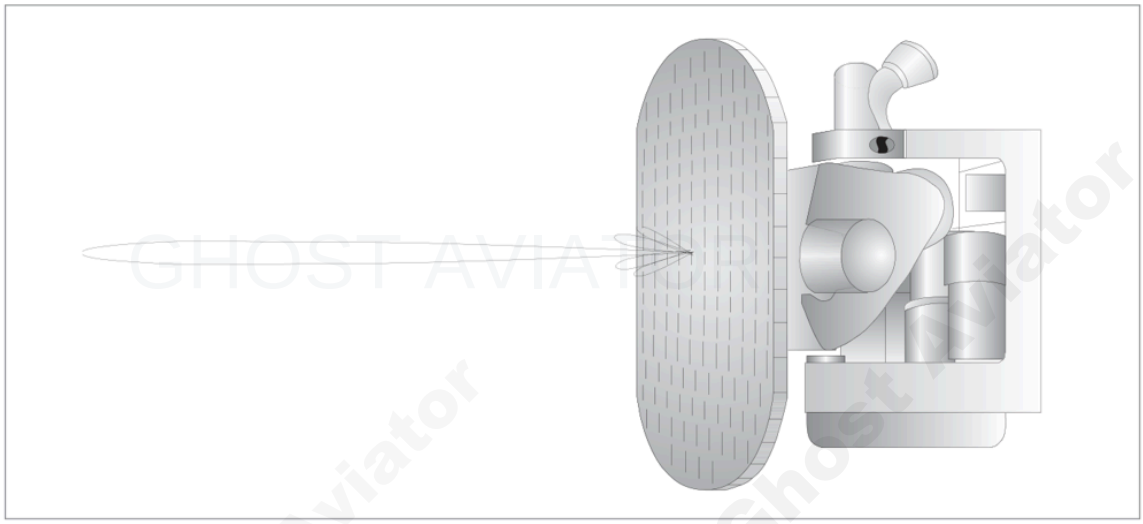


Figure 4.12 Phased array or slotted antenna

Figure 4.12 — Flat plate phased array / slotted antenna: a flat plate with numerous waveguide-sized slots. Each slot is fed RF energy from behind the plate and acts as an individual radiating element. The transmitted beam results from the interaction of all individual slot beams

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✅ **Five Advantages of Phased Array over Parabolic Reflector**

1. **Narrower beam** — better target discrimination / azimuth resolution
2. **Reduced side lobes** — fewer false returns, less interference
3. **Less power required** for a given range — more efficient RF transmission
4. **Narrower pulse** — better range resolution
5. **Improved resolution** — better target separation in range and azimuth

💡 **Exam Trap — Q3 in This Chapter**

Q3 asks which is **NOT** an advantage of slotted antenna over parabolic. The answer is **"Directivity" (d)** — *both* the parabolic dish and the phased array are highly directional. Directivity is a shared characteristic, not an advantage of the phased array specifically. The *advantages* are: narrower beam, fewer side lobes, less power, narrower pulse, improved resolution.

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⚡ QUICK REVISION SUMMARY — CHAPTER 4

- **Dipole:** centre-fed, λ/2 total; radiates perpendicular to aerial axis
- **Marconi:** λ/4 above ground plane; ground plane acts as second half; used on aircraft
- **Wavelength:** λ = 300/f(MHz); dipole = λ/2; Marconi = λ/4
- **Velocity factor:** actual length = **0.95 ×** theoretical length (5% shorter)
- **Feeders:** LF/MF = wire; HF/VHF = twin wire; UHF = coax; SHF+ = waveguide
- **Waveguide:** hollow rectangular tube; internal dimension = λ/2
- **Polar diagram:** dipole = torus (3D) / figure-eight (vertical section)
- **Reflector:** 5% longer than aerial, λ/4 behind, 180° out of phase → concentrates beam forward
- **Directors:** shorter than aerial, in front → narrow, focus beam; side lobes are a problem
- **ILS localizer:** 16 or 24 aerial array, λ/2 spacing → two narrow symmetrical lobes
- **ADF loop:** figure-of-eight polar diagram; two nulls used for DF; null sharper than maximum
- **Parabolic dish:** waveguide horn at focal point → parallel in-phase reflected rays → narrow beam + side lobes
- **Phased array advantages (5):** narrower beam, fewer side lobes, less power, narrower pulse, better resolution

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📝 Practice Questions & Detailed Answers

Q1. The ideal length for a Marconi aerial for a frequency of 406 MHz is:

a. 36.9 cm

b. 35.1 cm

c. 17.5 cm

d. 18.5 cm

✓ Correct Answer: **(c)** — 17.5 cm

Explanation:
λ = 300/406 = 0.739 m
Theoretical Marconi (λ/4) = 0.739/4 = 0.1847 m = 18.47 cm
Apply velocity factor: 0.95 × 18.47 = **17.55 cm ≈ 17.5 cm**
See Section 3.

Why the other options are wrong:

- **(a)** — 36.9 cm = λ/2 (theoretical half-wave dipole for 406 MHz). This is the dipole length without velocity factor correction.
- **(b)** — 35.1 cm = 0.95 × λ/2 = velocity-corrected dipole length. Wrong type of aerial (dipole, not Marconi).
- **(d)** — 18.5 cm = λ/4 (theoretical Marconi) without velocity factor. The question asks for the *ideal* length which requires the velocity factor correction to 17.5 cm.

Instructor's Note: *All four options are physically derived from 406 MHz. (a) = λ/2, (b) = 0.95λ/2, (c) = 0.95λ/4 ✓, (d) = λ/4. The question says "ideal" which means*

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applying the velocity factor. Knowing that the velocity correction is 0.95 is the key discriminator here.

Q2. A disadvantage of directivity is:

a. reduced range

b. side lobes

c. phase distortion

d. ambiguity

✓ Correct Answer: **(b)** — side lobes

Explanation: Achieving directivity through parasitic elements (reflectors and directors) inevitably produces **side lobes** — unwanted radiation/reception in directions other than the main beam. Side lobes can receive reflected signals (TV ghosting), cause interference in ILS and SSR systems, and produce false returns in radar. See Section 6.

Why the other options are wrong:

- **(a)** — Directivity typically *increases* range (by concentrating power into a beam), not reduces it. The opposite is true.
- **(c)** — Phase distortion is not a direct consequence of directivity in aerials. Phase shift is intentionally used to achieve directivity in phased arrays.
- **(d)** — Ambiguity is a characteristic of the ADF loop aerial (180° ambiguity from the two nulls), not of directivity in general.

Instructor's Note: *The chapter explicitly states: "directivity comes with its own price...many unwanted side lobes." This is the single stated disadvantage of directivity in the text. Side lobes are mentioned for TV aerials, SSR, ILS, and radar — making this an important concept across multiple systems.*

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Q3. Which of the following is **NOT** an advantage of a slotted antenna (phased array)?

a. Reduced side lobes

b. Improved resolution

c. Reduced power

d. Directivity

✓ Correct Answer: **(d)** — Directivity

Explanation: The five stated advantages of phased array over parabolic reflector are: narrow beam, reduced side lobes, less power, narrower pulse, improved resolution. **Directivity** is NOT listed as an advantage because *both* the parabolic dish and the phased array are already directional — directivity is not unique to phased arrays. Options (a), (b), and (c) are all genuine advantages of phased arrays. See Section 8.

Why the other options are incorrect (they ARE advantages):

- **(a)** — Reduced side lobes IS a phased array advantage. Fewer false radar returns.
- **(b)** — Improved resolution IS a phased array advantage. Better target discrimination.
- **(c)** — Reduced power IS a phased array advantage. Same range for less transmitted power.

Instructor's Note: *"NOT an advantage" questions require reading carefully. The answer is the one that both systems share equally — directivity. The parabolic dish is already extremely directional; the phased array doesn't gain directivity over the dish — it gains narrower beam, fewer side lobes, less power, narrower pulse, and better resolution.*

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Q4. The ideal length of a half-wave dipole for a frequency of 75 MHz is:

a. 1.9 m

b. 95 cm

c. 3.8 m

d. 47.5 cm

✓ Correct Answer: **(a)** — 1.9 m

Explanation:

λ = 300/75 = 4.0 m

Theoretical half-wave dipole (λ/2) = 4.0/2 = 2.0 m

Apply velocity factor: 0.95 × 2.0 = **1.9 m**

See Section 3.

Why the other options are wrong:

- **(b)** — 95 cm = 0.95 m. This is 0.95 × λ/4 = velocity-corrected Marconi (quarter-wave) length. Wrong aerial type.
- **(c)** — 3.8 m = 0.95 × λ = 0.95 × 4 m. No standard aerial type uses a full wavelength × 0.95.
- **(d)** — 47.5 cm = λ/4 × (some factor) — doesn't correspond to a standard calculation for this frequency.

Instructor's Note: *Same structure as Q1 but for a dipole (λ/2) at 75 MHz. Theoretical = 2.0 m; velocity-corrected = 1.9 m. Note that 95 cm (option b) is the Marconi length at this frequency — a deliberate trap to confuse dipole vs Marconi. Always confirm: dipole = λ/2; Marconi = λ/4.*

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📊 Master Reference Tables

Aerial Length Formula Summary

Aerial Type	Theoretical Length	Actual (×0.95)	Formula
Half-wave Dipole	λ/2	**0.95 × λ/2**	0.95 × 150/f(MHz) metres
Marconi (Quarter-wave)	λ/4	**0.95 × λ/4**	0.95 × 75/f(MHz) metres

Worked Calculation Reference

Frequency	λ (m)	Dipole (theo)	Dipole (actual)	Marconi (theo)	Marconi (actual)
75 MHz	4.0 m	2.0 m	**1.9 m**	1.0 m	**95 cm**
125 MHz	2.4 m	1.2 m	114 cm	60 cm	**57 cm**
406 MHz	0.739 m	36.9 cm	**35.1 cm**	18.5 cm	**17.5 cm**

Feeder Types by Frequency

Band	Feeder
LF, MF	Simple wire
HF, VHF	Twin wire feeder
UHF	Coaxial cable
Upper UHF, SHF, EHF	Waveguide (hollow rectangular tube; dim = λ/2)

Mnemonics & Memory Aids

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- **λ = 300/f(MHz)** — "300 divided by MHz gives metres"
- **Dipole = λ/2; Marconi = λ/4** — "Dipole = Double, Marconi = Mini"
- **Velocity factor = 0.95** — "95% of theory = actual ideal"
- **Feeder mnemonic:** "Wire → Twin → Coax → Waveguide" (low to high frequency)
- **Waveguide dimension = λ/2**
- **Reflector = 5% longer, λ/4 behind, 180° out of phase**
- **ADF null: sharper than max → better DF accuracy → but 180° ambiguity**
- **Phased array advantages (5):** "Narrow Beam, Fewer Side lobes, Less Power, Narrow Pulse, Better Resolution" → NB-FSL-NPR
- **Directivity is NOT an advantage of phased over parabolic** (both are directional)

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